Resource-dependent bounds on reciprocal absorption–scattering asymmetry

Full-vector Maxwell estimates, an exact current relaxation, and physical small-volume scaling

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Abstract

We establish a strictly positive resource-dependent gap below unit extinction-normalized directional absorption asymmetry for arbitrary compact reciprocal dielectric bodies in vacuum. The sufficient material resource is the integrated loss-weighted response $M = \int_D \eta$, where $|\chi v|^2 \le \eta v^\dagger (\mathrm{Im}_H \chi) v$. A direct all-angle energy estimate gives $\Gamma \le k^3 M/(6\pi + k^3 M)$. For bounded local response $\eta \le \bar{\eta}$ and support in a radius-a ball, the complete electric-current radiation spectrum satisfies $\|\mathcal{R}_a\| \le ka$ at every frequency, without a multipole cutoff. This improves the efficiency parameter to $\min\{k^3 M/(6\pi), \bar{\eta}\,\|\mathcal{R}_a\|\}$. At any positive required extinction a resolvent of this full radiation operator supplies an additional strictly positive scattering floor. Its value is the exact minimum of a stated current relaxation, proved by an explicit optimizer. A homogeneous, strictly lossy, reciprocal anisotropic sphere has the exact-Maxwell expansion $\Gamma = (160/2071)(ka)^3 + O((ka)^4)$, establishing the small-volume exponent for the allowed polarization class. The elementary response and efficiency ingredients have substantial prior art. The optimal fixed-resource Maxwell supremum, simultaneous attainability by one local material object, and the class with only a one-sided real permittivity cap remain unresolved. These distinctions prevent a claim that the original Open Question 7 is fully closed.

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1 Main statement, admissible class, and units

Miller and Monticone's OQ7 asks for fundamental bounds on the redistribution between absorption and scattering when a reciprocal object is illuminated by time-reversed waves with equal extinction [1]. The present work addresses one fixed, positive vacuum frequency. Write $k = \omega/c > 0$ and use the $e^{-\mathrm{i}\omega t}$ convention. All permittivities in this manuscript are *relative* permittivities. The background is vacuum and the relative permeability is I until Section 7.

Let $D \subset \mathbb{R}^3$ be a bounded measurable material support, with compact closure, and let $\chi = \varepsilon_r - I$ be a local, essentially bounded complex 3×3 susceptibility on D. Reciprocity means $\chi^T = \chi$ almost everywhere. The Hermitian real and imaginary parts are $\mathrm{Re}_H\, \chi = (\chi + \chi^\dagger)/2$ and $B = \mathrm{Im}_H\, \chi = (\chi - \chi^\dagger)/(2\mathrm{i})$. For a unit vector $d \in S^2$ and a complex unit polarization $e^\dagger e = 1$, $d \cdot e = 0$, the two electric incident fields are

$$f_+(x) = e\, e^{\mathrm{i}kd\cdot x}, \qquad f_-(x) = \overline{e}\, e^{-\mathrm{i}kd\cdot x} = \overline{f_+(x)}. \tag{1}$$

Their magnetic fields obey $Z_0 H_- = -\overline{Z_0 H_+}$. Conjugating the electric polarization matters for elliptical illumination. Reciprocity equates the extinction cross sections: $X = X_+ = X_- > 0$. Set

$$\Gamma = \frac{|A_+ - A_-|}{X} = \frac{|S_+ - S_-|}{X}. \tag{2}$$

Here A, S, X denote absorption, total scattering, and extinction cross sections, respectively; all have units of area. Total scattering includes both transverse polarizations and the entire sphere of radiation.

Theorem 1.1 (A closed fixed-resource bound)**.** *Fix $k > 0$, a maximum diameter $R > 0$, a volume cap $V > 0$, $P \geq 1$, and $0 < \delta \leq Q < \infty$. For every support D with $\operatorname{diam} D \leq R$ and $|D| \leq V$, and every essentially bounded reciprocal susceptibility satisfying*

$$I \preceq \mathrm{Re}_H\, \varepsilon_r(x) \preceq PI, \qquad \delta I \preceq B(x) \preceq QI \quad \text{a.e. on } D, \tag{3}$$

the outgoing Maxwell scattering problem is uniquely solvable in the volume-integral formulation described below. For every pair (1) with $X > 0$,

$$\boxed{\Gamma \leq \frac{b_0}{1+b_0} < 1, \qquad b_0 = \eta_0 \min\left\{\frac{k^3 V_{\mathrm{eff}}}{6\pi}, \sqrt{\frac{3}{8}}\, kR\right\}, \qquad \eta_0 = \frac{(P-1)^2}{\delta} + Q,} \tag{4}$$

where $V_{\mathrm{eff}} = \min\{V, \pi R^3/6\}$. Consequently the same result holds after imposing any additional minimum extinction $X \geq E_0 > 0$. No extinction floor is needed for this strict uniform gap.

The theorem's lower real-part assumption is explicit. More generally it is enough to assume $\|\mathrm{Re}_H\, \chi\| \leq c$ and replace η_0 by $c^2/\delta + Q$. If $0 \preceq \mathrm{Re}_H\, \varepsilon_r \preceq PI$, one may take $c = \max\{1, |P-1|\}$. An upper real-part bound by itself allows arbitrarily negative contrast and does not imply this response bound; Section 7.3 states precisely what is and is not known in that enlarged class.

For a dimensionally consistent version of the function in the research question, put $\kappa = kR$, $v = k^3 V$, and $e_0 = k^2 E_0$. One valid strictly positive function, on every nonempty class (3), is

$$F(\kappa, v, P, Q, \delta, e_0) = \left[1 + \left(\frac{(P-1)^2}{\delta} + Q\right) \min\left\{\frac{\min\{v, \pi\kappa^3/6\}}{6\pi}, \sqrt{\frac{3}{8}}\, \kappa\right\}\right]^{-1}. \tag{5}$$

The absence of e_0 from this sufficient expression is intentional. Section 5 gives a stronger positive-extinction bound. If one retains dimensional V and E_0 as arguments of F, k must also be fixed or supplied; kR alone does not nondimensionalize them.

2 Exact Maxwell identities and well-posedness

Divide all electric fields by the nonzero incident amplitude E_{ref} and set $p = \chi E$. Thus E and p in the normalized formulas are dimensionless, and the physical polarization density is $\varepsilon_0 E_{\mathrm{ref}} p$. The inner product $\langle u, v\rangle = \int \overline{u} \cdot v$ is conjugate-linear in its first argument. For $g_k(x) = e^{\mathrm{i}k|x|}/(4\pi\,|x|)$, define the outgoing distributional volume operator, restricted to D after zero extension,

$$\mathcal{T}_k p = (k^2 I + \nabla\nabla\cdot)(g_k * p), \qquad E = f + \mathcal{T}_k p. \tag{6}$$

The distributional electrostatic contact term is part of $\mathcal{T}_k$ throughout. Its far-field amplitude and exact cross sections are

$$F_p(n) = \frac{k^2}{4\pi}\Pi_n \int_D p(x) e^{-\mathrm{i}kn\cdot x}\,\mathrm{d}x, \qquad \Pi_n = I - nn^T, \tag{7}$$

$$X = k\,\mathrm{Im}\,\langle f, p\rangle\,, \qquad A = k\int_D E^\dagger B E\,\mathrm{d}x, \qquad S = \int_{S^2} |F_p(n)|^2\,\mathrm{d}\Omega, \tag{8}$$

$$X = A + S. \tag{9}$$

The forward optical theorem is $X = (4\pi/k)\,\mathrm{Im}(e^\dagger F_p(d))$. These normalizations follow by dividing physical powers by the incident intensity $|E_{\mathrm{ref}}|^2/(2Z_0)$.

Lemma 2.1 (Bounded full dyadic operator)**.** *Let $v_D = |D| < \infty$ and $a_v = (3v_D/(4\pi))^{1/3}$. On $L^2(D;\mathbb{C}^3)$,*

$$\|\mathcal{T}_0\| \le 1, \qquad \|\mathcal{T}_k - \mathcal{T}_0\| \le \frac{3}{4}k^2 a_v^2. \tag{10}$$

Proof. On all of $\mathbb{R}^3$, the Fourier multiplier of $\mathcal{T}_0$ is $-\xi\xi^T/|\xi|^2$ almost everywhere. Zero extension and restriction are contractions, so the first bound follows, including the contact term. The difference has an ordinary locally integrable kernel. Put $r = |x|$, $u = kr$, and $b(u) = e^{\mathrm{i}u}(\mathrm{i}u - 1) + 1$. Its radial and tangential eigenvalues are, respectively,

$$\frac{-2b(u)}{4\pi r^3}, \qquad \frac{u^2 e^{\mathrm{i}u} + b(u)}{4\pi r^3}.$$

Since $b(0) = 0$ and $b'(u) = -ue^{\mathrm{i}u}$, $|b(u)| \le u^2/2$. The kernel norm is therefore at most $3k^2/(8\pi r)$. Radial rearrangement of a decreasing function gives $\sup_x \int_D |x-y|^{-1}\,\mathrm{d}y \le 2\pi a_v^2$; the same estimate holds with x and y interchanged. Schur's test gives (10). □

The imaginary part of the outgoing Green operator is positive:

$$\langle p, (\mathrm{Im}_H\,\mathcal{T}_k)p\rangle = \frac{k^3}{16\pi^2}\int_{S^2}\left|\Pi_n\int_D p(x)e^{-\mathrm{i}kn\cdot x}\,\mathrm{d}x\right|^2\mathrm{d}\Omega = S/k. \tag{11}$$

Indeed, $\sin(kr)/(4\pi r) = k(16\pi^2)^{-1}\int_{S^2} e^{\mathrm{i}kn\cdot x}\,\mathrm{d}\Omega$; applying $k^2 I + \nabla\nabla\cdot$ gives the transverse plane-wave representation in (11). Density extends the identity from smooth compact currents to $L^2(D)$.

Proposition 2.2 (Coercivity and reciprocal extinction)**.** *If $B \succeq \delta I > 0$ and $\|\chi\| \le K < \infty$ a.e., then (6) has a unique solution $p \in L^2(D;\mathbb{C}^3)$. If also $\chi^T = \chi$, the pair (1) has equal extinction.*

Proof. The susceptibility is invertible pointwise and its inverse is bounded by $1/\delta$. Set $W = -\,\mathrm{Im}_H\,\chi^{-1} = \chi^{-\dagger}B\chi^{-1}$. For every v, $v^\dagger W v \ge \delta\,|\chi^{-1}v|^2 \ge (\delta/K^2)\,|v|^2$. For $\mathcal{L} = \chi^{-1} - \mathcal{T}_k$,

$$-\,\mathrm{Im}\,\langle p, \mathcal{L}p\rangle = \langle p, Wp\rangle + \langle p, (\mathrm{Im}_H\,\mathcal{T}_k)p\rangle \ge (\delta/K^2)\,\|p\|_2^2\,.$$

The bounded complex sesquilinear form obtained by multiplication by i is coercive. The complex Lax–Milgram theorem yields a bounded inverse and a unique $p = \mathcal{L}^{-1} f$. The constructed field is in $H_{\mathrm{loc}}(\mathrm{curl})$: the scalar Helmholtz volume potential is locally H^2, and $\nabla \times E = \nabla \times f + k^2 \nabla \times (g_k * p)$ is locally H^1. It solves Maxwell's equations distributionally and satisfies the outgoing radiation condition. Conversely, an outgoing finite-energy weak Maxwell field has this Green representation: subtracting the volume potential leaves an entire outgoing source-free vacuum Maxwell field, which vanishes by the vacuum uniqueness theorem. The same energy identity with $\mathcal{L}p = f$ gives (9).

The kernel of $\mathcal{T}_k$ is symmetric under simultaneous transpose and interchange of its two arguments. Consequently $\mathcal{L}$ and $\mathcal{L}^{-1}$ are symmetric for the continuous bilinear pairing $[u, v] = \int u \cdot v$, when $\chi^T = \chi$. Since $f_- = \overline{f_+}$,

$$\langle f_+, p_+ \rangle = [f_-, \mathcal{L}^{-1} f_+] = [f_+, \mathcal{L}^{-1} f_-] = \langle f_-, p_- \rangle .$$

Taking imaginary parts proves $X_+ = X_-$. □

This proof does not require a smooth boundary, a finite number of components, a minimum feature size, or a modal truncation. It does require the full tensor loss floor when used to establish existence for the entire essentially bounded coefficient class.

3 The response budget and an all-angle efficiency bound

A broader hypothesis sufficient for the elementary bounds is

$$\begin{gathered} \eta : D \to [0, \infty) \text{ measurable}, \qquad B(x) \succeq 0, \\ |\chi(x) v|^2 \le \eta(x) v^\dagger B(x) v \quad \text{for all } v \in \mathbb{C}^3, \qquad 0 < M := \int_D \eta(x) \, \mathrm{d}x < \infty. \end{gathered} \tag{12}$$

Here η is dimensionless and M has units of volume. This is a loss-weighted material-response budget, not the integral of loss, not absorbed power, and not a frequency sum rule. In this broader class the statements apply to every well-posed finite-absorption Maxwell solution; (12) alone is not asserted to establish existence for arbitrary rough coefficients.

When $B \succ 0$ the smallest admissible value is $\eta = \left\| \chi B^{-1} \chi^\dagger \right\|$. For singular B this becomes $\left\| \chi B^{+} \chi^\dagger \right\|$ provided $\ker B \subseteq \ker \chi$; without that kernel condition the pseudoinverse expression is insufficient. The convention $\eta = 0$, $p = 0$ on nonresponding material is harmless. Writing $\chi = A_0 + \mathrm{i}B$ with $A_0 = \mathrm{Re}_H \, \chi$ gives the exact noncommuting-matrix identity

$$\chi B^{-1} \chi^\dagger = A_0 B^{-1} A_0 + B. \tag{13}$$

Thus $\|A_0\| \le c$ and $\delta I \preceq B \preceq Q I$ imply $\eta \le c^2/\delta + Q$.

Theorem 3.1 (Integrated response bound)**.** *Under* (12)*, each unit-amplitude plane-wave solution satisfies*

$$S \le \beta_M A, \qquad X^2 \le kMA, \qquad X \le kM, \qquad \beta_M = \frac{k^3 M}{6\pi}. \tag{14}$$

For a reciprocal pair,

$$\Gamma \le \min \left\{ \frac{\beta_M}{1 + \beta_M}, \, 1 - \frac{X}{kM} \right\}. \tag{15}$$

Proof. The material condition implies $\int_D |p|^2/\eta \le A/k$, with zero integrand where $\eta = 0$. Weighted Cauchy–Schwarz yields $\|p\|_1^2 \le MA/k$. This proof only needs $p \in L^1$; it does not infer $p \in L^2$ from $\eta \in L^1$. Minkowski's inequality, followed by $\int_{S^2} \Pi_n \,\mathrm{d}\Omega = (8\pi/3)I$, gives

$$S^{1/2} \le \frac{k^2}{4\pi} \int_D \left(\int_{S^2} |\Pi_n p(x)|^2 \,\mathrm{d}\Omega \right)^{1/2} \mathrm{d}x = \frac{k^2}{4\pi} \sqrt{\frac{8\pi}{3}} \, \|p\|_1 .$$

This proves the first inequality in (14). Because $|f| = 1$, $|\langle f, p\rangle| \le \|p\|_1$, so $X^2 \le k^2 \|p\|_1^2 \le kMA$. Since $A \le X$, one obtains $X \le kM$ for $X > 0$. For $s = S/X$, the first two bounds give $0 \le s \le \min\{\beta_M/(1+\beta_M), 1 - X/(kM)\}$. Both reciprocal illuminations lie in that interval; its width bounds $\Gamma = |s_+ - s_-|$. □

The response factor and radiation-efficiency principle have direct antecedents in material-response bounds and current optimization [2, 3, 4, 5]. In particular, Schab et al. explicitly identify the scattering–absorption boundary using the largest radiation-to-loss generalized eigenvalue [3, Sec. 3.1]. The elementary asymmetry bound is a consequence of that established principle and reciprocal extinction equality.

Corollary 3.2 (Necessary costs and absolute differences)**.** *If $\Gamma \ge 1 - \epsilon$, where $0 < \epsilon < 1$, then*

$$M \ge \frac{6\pi}{k^3} \frac{1-\epsilon}{\epsilon}. \tag{16}$$

If also $X \ge E_0 > 0$, then $M \ge E_0/(k\epsilon)$. For the absolute absorption or scattering difference $\Delta = X\Gamma$,

$$\Delta \le \begin{cases} kM\,\beta_M/(1+\beta_M)^2, & \beta_M \le 1, \\ kM/4, & \beta_M \ge 1. \end{cases} \tag{17}$$

Proof. Rearrange (15). For the last statement put $x = X/(kM) \in [0,1]$ and $r = \beta_M/(1+\beta_M)$; then $\Delta/(kM) \le \min\{rx, x(1-x)\}$. If $r \le 1/2$ the maximum is at $x = 1 - r$ and equals $r(1-r)$. If $r \ge 1/2$ it is at $x = 1/2$ and equals $1/4$. □

4 The complete ball radiation spectrum and a linear size estimate

Assume now $D \subset B_a$ and $0 < \eta(x) \le \bar{\eta} < \infty$ on responding material. Extend p by zero to B_a. The material condition then supplies $\|p\|_2^2 \le \bar{\eta} A/k$. Let

$$(\mathcal{H}p)(n) = \Pi_n \int_{B_a} p(x) e^{-\mathrm{i}kn\cdot x} \,\mathrm{d}x, \qquad \mathcal{R}_a = \mathrm{Im}_H \, \mathcal{T}_k|_{B_a} = \frac{k^3}{16\pi^2} \mathcal{H}^\dagger \mathcal{H}.$$

The operator $\mathcal{R}_a$ is positive, compact, and self-adjoint. Its units and eigenvalues are dimensionless.

Lemma 4.1 (All vector radiation eigenvalues)**.** *For $z = ka$ the positive eigenvalues of $\mathcal{R}_a$ are*

$$\lambda_\ell^M(z) = \int_0^z s^2 j_\ell(s)^2 \,\mathrm{d}s, \tag{18}$$

$$\lambda_\ell^N(z) = \int_0^z \left[\ell(\ell+1) j_\ell(s)^2 + \big((s j_\ell(s))'\big)^2 \right] \mathrm{d}s, \qquad \ell = 1, 2, \ldots. \tag{19}$$

Each family has multiplicity $2\ell + 1$; coincidences add multiplicities. The orthogonal complement of these radiating modes is $\ker \mathcal{R}_a$. For the plane wave $f = e e^{\mathrm{i}kd\cdot x}$ on B_a and $\alpha > 0$,

$$D_\alpha := \left\langle f, (\mathcal{R}_a + \alpha I)^{-1} f \right\rangle = \frac{2\pi}{k^3} \sum_{\ell=1}^{\infty} (2\ell+1) \left[\frac{\lambda_\ell^M}{\lambda_\ell^M + \alpha} + \frac{\lambda_\ell^N}{\lambda_\ell^N + \alpha} \right]. \tag{20}$$

In particular,

$$\frac{2\pi}{k^3}\sum_{\ell=1}^{\infty}(2\ell+1)(\lambda_\ell^M+\lambda_\ell^N)=\|f\|_2^2=\frac{4\pi a^3}{3}. \tag{21}$$

Proof. Take scalar spherical harmonics $Y_{\ell m}$ normalized on S^2 and the associated orthonormal tangent vector harmonics $X_{\ell m}=-in\times\nabla_{S^2}Y_{\ell m}/\sqrt{\ell(\ell+1)}$ and $n\times X_{\ell m}$. The Rayleigh plane-wave expansion and angular integration give

$$\mathcal{H}^\dagger X_{\ell m}(r)=4\pi i^\ell j_\ell(k\,|r|)X_{\ell m}(\widehat{r}),\qquad \mathcal{H}^\dagger(n\times X_{\ell m})=(ik)^{-1}\nabla\times\mathcal{H}^\dagger X_{\ell m}.$$

Thus the adjoint sends these to 4π, up to unit phases, times the regular magnetic and electric vector spherical waves. For the magnetic wave its squared radial integral is $k^{-3}\int_0^z s^2 j_\ell(s)^2\,\mathrm{d}s$. For the electric wave, the orthogonal radial and tangential parts give $k^{-3}\int_0^z[\ell(\ell+1)j_\ell(s)^2+((sj_\ell(s))')^2]\,\mathrm{d}s$. Different modes are orthogonal on B_a. Multiplication by $k^3/(16\pi^2)$ in $\mathcal{R}_a$ gives exactly (18)–(19). Completeness of the two families of tangent harmonics gives all nonzero singular values of $\mathcal{H}$, hence all positive eigenvalues of $\mathcal{R}_a$.

If $q\in\ker\mathcal{R}_a$, then $\mathcal{H}q=0$ almost everywhere. Because B_a has finite volume, $q\in L^1$ and $\mathcal{H}q$ is continuous; therefore it vanishes at d. It follows that $\langle f,q\rangle=e^\dagger(\mathcal{H}q)(d)=0$. Thus $f\perp\ker\mathcal{R}_a$; no angular delta distribution is treated as an L^2 vector. The vector addition theorem gives $\sum_m\left|e^\dagger X_{\ell m}(d)\right|^2=(2\ell+1)/(8\pi)$, and the same identity for the other tangent family. For an angular eigenmode u with eigenvalue λ the normalized volume mode is $q=k^{3/2}\mathcal{H}^\dagger u/(4\pi\sqrt{\lambda})$. Consequently $\mathcal{H}q=4\pi k^{-3/2}\sqrt{\lambda}\,u$. The squared plane-wave projection summed over m is therefore $(2\pi/k^3)(2\ell+1)\lambda$ for each family. The spectral theorem proves (20) and (21). □

Spherical radiation modes and their use in material bounds are established tools [4, 5, 6, 17]. The next elementary estimate is useful here because it is uniform over all angular orders.

Theorem 4.2 (Uniform linear bound, with no cutoff)**.** *For every $z>0$,*

$$\Lambda(z):=\|\mathcal{R}_a\|=\max_{\ell\ge1}\{\lambda_\ell^M(z),\lambda_\ell^N(z)\}\le\min\left\{\frac{2z^3}{9},\,z\right\}. \tag{22}$$

Hence

$$S\le\beta A,\qquad \beta=\min\left\{\frac{k^3M}{6\pi},\,\bar{\eta}\Lambda(ka)\right\}. \tag{23}$$

Proof. Apply the all-angle L^1 estimate of Theorem 3.1 to an arbitrary current in B_a, then use $\|p\|_1^2\le|B_a|\,\|p\|_2^2$. Since $S/k=\langle p,\mathcal{R}_a p\rangle$, this gives $\Lambda(z)\le k^3\,|B_a|\,/(6\pi)=2z^3/9$.

For the linear estimate fix any integer $\ell\ge1$ and put $u(z)=zj_\ell(z)$, $L=\ell(\ell+1)$, and $\mathcal{E}_\ell=u'^2+(1-L/z^2)u^2$. The Bessel equation gives

$$u''+(1-L/z^2)u=0,\qquad \mathcal{E}_\ell'(z)=2Lu(z)^2/z^3\ge0.$$

For this fixed ℓ, the standard sine and cosine asymptotics yield $\mathcal{E}_\ell(z)\to1$ as $z\to\infty$. Thus $\mathcal{E}_\ell(z)\le1$ for every $z>0$ and every ℓ. Direct differentiation and the zero limits at $z=0$ show

$$\lambda_\ell^M=\frac{z\mathcal{E}_\ell-uu'}{2},\qquad \lambda_\ell^N=\frac{z\mathcal{E}_\ell+uu'}{2}.$$

Both are nonnegative by their integral definitions and their sum is $z\mathcal{E}_\ell\le z$. Taking the supremum proves the bound. This argument proves the estimate separately for every ℓ and then takes a supremum; it does not interchange a nonuniform asymptotic limit with an infinite supremum. Finally $S=k\,\langle p,\mathcal{R}_a p\rangle\le k\Lambda\,\|p\|_2^2\le\bar{\eta}\Lambda A$, which combines with Theorem 3.1 to give (23). □

The linear order is sharp for the radiation *operator*. For $\ell = 1$,

$$\mathcal{E}_1(z) = 1 - z^{-2} + \sin(2z)z^{-3} - \sin^2(z)z^{-4},$$

so the larger of the two eigenvalues is at least

$$\frac{z\mathcal{E}_1(z)}{2} \geq \frac{z}{2} - \frac{1}{2z} - \frac{1}{2z^2} - \frac{1}{2z^3}.$$

Together with (22), this proves $\Lambda(z) = \Theta(z)$ as $z \to \infty$. It does not prove $\Lambda \sim z/2$ or a matching physical asymmetry construction. Equation (23) also implies the necessary size cost

$$\Gamma \geq 1 - \epsilon \quad \Longrightarrow \quad \bar{\eta}\, ka \geq \frac{1-\epsilon}{\epsilon}. \tag{24}$$

Completion of Theorem 1.1. Equation (13) gives $\bar{\eta} \leq \eta_0$ and $M \leq \eta_0 |D|$. The isodiametric inequality gives $|D| \leq \pi R^3/6$. For completeness, from a radius-a smallest enclosing ball one can select at most four contact points y_i and nonnegative weights μ_i with $\sum_i \mu_i = 1$, $\sum_i \mu_i y_i = 0$, and $|y_i| = a$, after centering. The center lies in the convex hull of the contact set, since otherwise a separating hyperplane would permit moving it to decrease the maximal distance. Carathéodory's theorem supplies four contact points. Consequently

$$2a^2 = \sum_{i,j} \mu_i \mu_j \, |y_i - y_j|^2 \leq R^2 \left(1 - \sum_i \mu_i^2\right) \leq \frac{3}{4} R^2.$$

Thus $a \leq \sqrt{3/8}\, R$. Using (22) in (23) gives $\beta \leq b_0$, and the interval argument of Theorem 3.1 proves (4). The susceptibility is bounded under (3), so Proposition 2.2 supplies well-posedness. □

An arbitrary diameter-R body need not fit in a radius-$R/2$ ball. The volume inequality and enclosing-ball inequality used above are different statements. Taking $a = R$ would also be valid, but weaker.

5 Positive extinction: an exact all-mode current relaxation

The optical theorem alone controls a forward amplitude, whereas S is an angular integral. Bounding their ratio by a heuristic directivity or by a finite multipole count would leave a superdirectivity loophole. The following argument instead retains the entire radiation operator and the loss-controlled norm of the polarization. Let $a > 0$, $0 < \bar{\eta} < \infty$, $w = 1/\bar{\eta}$, and $t = X/k$. The quantity t and the function D_α in (20) have units of volume; α, w are dimensionless. Every physical current extended to B_a obeys

$$\operatorname{Im} \langle f, p \rangle = t, \qquad \langle p, (\mathcal{R}_a + wI)p \rangle \leq t. \tag{25}$$

The first equality is the optical theorem, and the second follows from $w \, \|p\|_2^2 \leq A/k$ and $X = A + S$.

Theorem 5.1 (Strict scattering floor and its exact relaxed value)**.** *For $0 < t \leq D_w$, define*

$$\rho_{a,w}(t) = \sup_{0<\alpha<w} \frac{wt/D_\alpha - \alpha}{w - \alpha}. \tag{26}$$

This function is nondecreasing in t and strictly positive for $t > 0$. It is the exact minimum of

$$\frac{\langle p, \mathcal{R}_a p \rangle}{t} \quad \text{over } p \in L^2(B_a; \mathbb{C}^3) \quad \text{subject to (25)}. \tag{27}$$

The feasible set is empty for $t > D_w$. In particular every physical solution satisfies $S/X \geq \rho_{a,w}(X/k)$.

Proof. For $0 < \alpha < w$, Cauchy–Schwarz in the positive metric $\mathcal{R}_a + \alpha I$ gives

$$\begin{aligned} t^2 \le |\langle f, p\rangle|^2 &\le D_\alpha \, \langle p, (\mathcal{R}_a + \alpha I)p\rangle \\ &\le D_\alpha \left[\left(1 - \frac{\alpha}{w}\right)\langle p, \mathcal{R}_a p\rangle + \frac{\alpha}{w} t\right]. \end{aligned}$$

Rearranging proves the lower bound (26). With $\alpha = w$ it gives $t \le D_w$. Every term in the supremum is increasing in t.

By Lemma 4.1, $f \perp \ker \mathcal{R}_a$. The spectral theorem and dominated convergence therefore give

$$\alpha D_\alpha = \left\langle f, \alpha(\mathcal{R}_a + \alpha I)^{-1} f \right\rangle \longrightarrow 0 \quad (\alpha \downarrow 0).$$

For any fixed $t > 0$ choose a sufficiently small α so that $\alpha D_\alpha < wt$. The corresponding quotient in (26) is positive. This proves strict positivity, with no finite-rank assumption.

It remains to prove attainment and exactness. Put

$$\begin{aligned} H_\alpha &= \left\langle f, (\mathcal{R}_a + \alpha I)^{-2} f \right\rangle = -D'_\alpha, \qquad K_\alpha = \left\langle f, (\mathcal{R}_a + \alpha I)^{-3} f \right\rangle, \\ J_\alpha &= D_\alpha + (w - \alpha) H_\alpha, \qquad\qquad T(\alpha) = D_\alpha^2 / J_\alpha. \end{aligned}$$

Differentiation, using $H'_\alpha = -2K_\alpha$, gives

$$T'(\alpha) = \frac{2D_\alpha (w - \alpha)(D_\alpha K_\alpha - H_\alpha^2)}{J_\alpha^2} > 0. \tag{28}$$

Cauchy–Schwarz proves nonnegativity of the numerator. It is strictly positive since f has nonzero aggregate projections onto every positive magnetic and electric order, and hence onto infinitely many distinct positive eigenvalues of the compact operator. The equality case of Cauchy–Schwarz would require support on a single eigenvalue.

As $\alpha \uparrow w$, $T(\alpha) \to D_w$. As $\alpha \downarrow 0$, the positive series (20) gives $D_\alpha \to \infty$. Also $H_\alpha \ge D_\alpha^2 / \|f\|_2^2 \to \infty$. The unit vectors $v_\alpha = (\mathcal{R}_a + \alpha I)^{-1} f / \sqrt{H_\alpha}$ converge weakly to zero: their coordinate along each fixed positive eigenvector tends to zero, their kernel coordinates vanish, and the spectral eigenvectors together with the kernel span the Hilbert space. Approximation by a finite span then proves weak convergence for every test vector. In particular $D_\alpha / \sqrt{H_\alpha} = \langle f, v_\alpha\rangle \to 0$. Since $w - \alpha$ stays positive near zero, $T(\alpha) \to 0$.

For every $0 < t < D_w$ there is therefore a unique $\alpha \in (0, w)$ with $T(\alpha) = t$. Define

$$p_\alpha = \mathrm{i} \frac{t}{D_\alpha} (\mathcal{R}_a + \alpha I)^{-1} f. \tag{29}$$

Direct substitution yields $\operatorname{Im}\langle f, p_\alpha\rangle = t$ and $\langle p_\alpha, (\mathcal{R}_a + wI)p_\alpha\rangle = t^2 J_\alpha / D_\alpha^2 = t$. The Cauchy–Schwarz step is an equality for this current, so it attains the lower bound. Its value is

$$\rho_{a,w}(t) = \frac{D_\alpha - \alpha H_\alpha}{D_\alpha + (w - \alpha) H_\alpha}, \qquad t = \frac{D_\alpha^2}{D_\alpha + (w - \alpha) H_\alpha}. \tag{30}$$

At $t = D_w$, equality in the w-metric Cauchy–Schwarz inequality forces the unique feasible current $p = \mathrm{i}(\mathcal{R}_a + wI)^{-1} f$. Its scattering fraction is $1 - wH_w / D_w$. This is the limit of (26) as $\alpha \uparrow w$. The endpoint $t = 0$ may be assigned $\rho(0) = 0$. □

Corollary 5.2 (Uniform positive-extinction asymmetry bound)**.** *Suppose $D \subset B_a$, $\eta \le \bar{\eta}$, $M \le M_0 < \infty$, and the reciprocal pair has $X \ge E_0 > 0$. Set*

$$\beta_0 = \min\{k^3 M_0/(6\pi),\, \bar{\eta}\Lambda(ka)\}.$$

For a nonempty admissible class,

$$\boxed{\Gamma \le \min\left\{\frac{\beta_0}{1+\beta_0}, 1 - \frac{E_0}{kM_0}\right\} - \rho_{a,1/\bar{\eta}}(E_0/k).} \tag{31}$$

In particular $E_0 \le kD_{1/\bar{\eta}}$ is necessary for feasibility. A negative right-hand side proves that the stated class is empty.

Proof. For each illumination the scattering fraction is at least $\rho(X/k) \ge \rho(E_0/k)$. Theorem 3.1 and (23) bound it above by the minimum in (31), after replacing M by M_0 and X by E_0 in the monotone directions. The width of this common interval bounds the difference. □

For the class in Theorem 1.1, take $a = \sqrt{3/8}\,R$, $\bar{\eta} = \eta_0$, and $M_0 = \eta_0 V_{\text{eff}}$. The ball and material bounds thus define a completely specified function of the dimensionless resources. A smaller known enclosing ball or a tighter integrated response budget improves it.

5.1 A closed angular-cap correction

There is also a simple explicit correction that only needs $M < \infty$ and an enclosing radius. It can be combined with (31).

Proposition 5.3 (Forward cap bound)**.** *For every solution with $D \subset B_a$, $M < \infty$, and $X > 0$,*

$$SA \ge \frac{kX^4}{96\pi M(1+ka)^2}. \tag{32}$$

Proof. Put $g(n) = e^{\dagger} F_p(n)$ and $h = kX/(4\pi) > 0$. The optical theorem gives $|g(d)| \ge h$. The inequalities $\|\Pi_n - \Pi_d\| = |\sin\theta| \le |n-d|$ and $\left|e^{-\mathrm{i}kn\cdot x} - e^{-\mathrm{i}kd\cdot x}\right| \le ka\,|n-d|$ imply

$$|g(n) - g(d)| \le L\,|n-d|\,, \qquad L = \frac{k^2}{4\pi}(1+ka)\sqrt{MA/k}.$$

Coordinates are centered on the enclosing ball and the incident amplitude is rephased to one there. This leaves all cross sections and the forward optical theorem in the stated normalization unchanged. Moreover $h \le k^2 \|p\|_1/(4\pi)$, so $h/L \le 1/(1+ka) \le 1$. Using chord distance $r = |n-d|$, the sphere area element of a radial cap is $2\pi r\,\mathrm{d}r$. Since $|F_p(n)| \ge |g(n)|$,

$$S \ge 2\pi \int_0^{h/L} r(h - Lr)^2\,\mathrm{d}r = \frac{\pi h^4}{6L^2} = \frac{kX^4}{96\pi MA(1+ka)^2}.$$

□

For fixed resources let

$$C_0 = \frac{kE_0^2}{96\pi M_0(1+ka)^2}, \qquad r_0 = \frac{1-\sqrt{1-4C_0}}{2}. \tag{33}$$

If $C_0 > 1/4$, there is no feasible object. Otherwise each scattering fraction belongs to $[r_0, 1 - r_0]$. Before using ρ, also check $E_0 \le kD_{1/\bar{\eta}}$; failure means the class is empty and ρ need not be evaluated. Intersecting all proved intervals gives the strongest combined envelope established in this manuscript:

$$\boxed{\Gamma \le U_0 - L_0, \qquad \begin{aligned} U_0 &= \min\left\{\frac{\beta_0}{1+\beta_0}, 1 - \frac{E_0}{kM_0}, 1 - r_0\right\}, \\ L_0 &= \max\left\{\rho_{a,1/\bar{\eta}}(E_0/k),\, r_0\right\}. \end{aligned}} \tag{34}$$

If $U_0 < L_0$, the resources are incompatible with positive extinction. The cap coefficient is not claimed optimal. The resolvent result is exact only for (25)–(27), not for the combined physical asymmetry problem.

5.2 What these proofs do about resonances and superdirectivity

All positive radiation eigenvalues, including the arbitrarily small high-order values, remain in (20). Superdirective currents have small radiation relative to their norm. The loss metric charges that norm in (25), and the optical theorem fixes the forward work at positive extinction. These are the quantitative restrictions behind $\rho > 0$. The bounded-loss Maxwell inverse in Proposition 2.2 also controls resonant amplification for the stated class. No assumption of a finite number of contributing multipoles, no angular cone approximation, and no exclusion of grazing radiation appears in these arguments. This complements established directivity–efficiency and antenna absorption results [7, 8].

6 A physical Maxwell construction with sharp small-volume order

The current optimizer (29) need not be generated by a local reciprocal susceptibility, much less by the same susceptibility under the opposite wave. A distinct physical construction can, however, settle one scaling exponent.

Theorem 6.1 (Exact small-sphere asymmetry)**.** *Let $D = B_a$ be a homogeneous sphere and fix*

$$\chi = A_0 + \mathrm{i}B, \qquad A_0 = \operatorname{diag}(1,2,1), \qquad B = \begin{pmatrix} 1/2 & 1/5 & 0 \\ 1/5 & 1/2 & 0 \\ 0 & 0 & 1/2 \end{pmatrix}. \tag{35}$$

Illuminate it with $d = (0,0,1)^T$, $e = (1,\mathrm{i},0)^T/\sqrt{2}$, and the time-reversed pair (1). For the exact outgoing Maxwell solution, as $ka \downarrow 0$ at fixed k,

$$\boxed{\Gamma_a = \frac{160}{2071}(ka)^3 + O((ka)^4) = \frac{120}{2071\pi}k^3\,|B_a| + O\Big(k^3\,|B_a|\;ka\Big).} \tag{36}$$

The material eigenvalue bounds are fixed: $\mathrm{Re}_H\,\varepsilon_r$ *has eigenvalues* $2, 3, 2$ *and* B *has eigenvalues* $0.3, 0.5, 0.7$.

Proof. Rescale $x = ay$ to the unit sphere and write $\kappa = ka$. The equation becomes $(\chi^{-1} - \mathcal{T}_\kappa)p_\kappa = f_\kappa$. Lemma 2.1 gives $\|\mathcal{T}_\kappa - \mathcal{T}_0\| \le 3\kappa^2/4$. The loss coercivity in Proposition 2.2 gives a uniform inverse bound as $\kappa \to 0$, because the fixed tensor (35) has a positive loss floor. For a constant vector in a ball, $\mathcal{T}_0 p = -p/3$ inside the ball. Thus the static solution is the constant vector Le, where

$$L = 3\chi(\chi + 3I)^{-1}.$$

The incident wave satisfies $f_\kappa = e + O(\kappa)$ in $L^2(B_1)$. The resolvent identity and the operator estimate imply $p_\kappa = Le + O(\kappa)$ in $L^2(B_1)$, and similarly for $\overline{e}$. All constants depend only on the displayed fixed tensor. After returning to physical coordinates, (7) then gives uniformly in $n \in S^2$, with $v_a = |B_a|$,

$$F_\pm(n) = \frac{k^2 v_a}{4\pi} \Pi_n L e_\pm + O(k^2 v_a \kappa), \qquad e_+ = e, \quad e_- = \overline{e}.$$

The estimate uses $L^2(B_1) \subset L^1(B_1)$ and $e^{-\mathrm{i}\kappa n\cdot y} = 1 + O(\kappa)$ uniformly on $B_1 \times S^2$. Integrating the complete far field and using the optical work gives

$$S_\pm = \frac{k^4 v_a^2}{6\pi} \|Le_\pm\|^2 + O(k^4 v_a^2 \kappa),$$
$$X = k v_a \operatorname{Im}(e^\dagger L e) + O(k v_a \kappa).$$

The equality of the two exact extinctions follows independently from reciprocity.

For explicit verification, put $d_0 = (4 + \mathrm{i}/2)(5 + \mathrm{i}/2) + 1/25 = (1979 + 450\mathrm{i})/100$. The transverse entries of L are

$$L_{11} = 3 - \frac{9(5 + \mathrm{i}/2)}{d_0}, \qquad L_{22} = 3 - \frac{9(4 + \mathrm{i}/2)}{d_0}, \qquad L_{12} = L_{21} = \frac{9\mathrm{i}}{5 d_0}.$$

Direct rational algebra gives

$$\operatorname{Im}(e^\dagger L e) = \frac{931950}{4118941} > 0, \qquad \|Le\|^2 - \|L\overline{e}\|^2 = \frac{324000}{4118941}.$$

Their ratio is $720/2071$. Divide $|S_+ - S_-|$ by X, then use $v_a = 4\pi a^3/3$ to obtain (36). □

This is an asymptotic theorem about the exact Maxwell solution, not an assumption that only dipoles exist. The higher multipoles are included in the uniformly controlled error. Since Γ_a is a positive constant times $k^3 v_a$, the general small-volume exponent in Theorem 3.1 cannot be improved uniformly over this polarization and anisotropy class. No optimality is claimed for the uniform coefficient. Nor does this family establish sharpness at fixed $E_0 > 0$, because $X_a \to 0$. For real linear polarization a homogeneous centrosymmetric sphere has equal opposite-direction absorption; the present example specifically uses the allowed complex time-reversed polarization pair. The underlying role of nonorthogonal modes in reciprocal anisotropic or inhomogeneous lossy media has precedent [10, 9].

The tensor value can be embedded in a passive causal frequency response. For example, with $\Omega > \omega_0$ and $\gamma > 0$,

$$\chi(\omega) = \frac{(\Omega^2 - \omega_0^2) A_0}{\Omega^2 - \omega^2} + \frac{\gamma \omega_0 B}{\omega_0^2 - \omega^2 - \mathrm{i}\gamma\omega},$$

where the first term is interpreted as the lossless causal oscillator limit, satisfies $\chi(\omega_0) = A_0 + \mathrm{i}B$. Both oscillator-strength tensors are positive semidefinite. This observation establishes compatibility of the specified single-frequency tensor with passivity and causality; it supplies no finite-bandwidth asymmetry theorem.

7 Broader physical classes and unresolved hypotheses

7.1 Isotropic and anisotropic dielectrics

The response theorem applies to arbitrary local reciprocal anisotropic tensors and, as a special case, isotropic scalar media. For a scalar susceptibility with $\operatorname{Im}\chi > 0$, $\eta = |\chi|^2 / \operatorname{Im}\chi$. No restriction to

layers, a scalar wave equation, a principal polarization, or finitely many spatial regions was used. The physical exponent construction in Section 6 is anisotropic. We have not supplied a matching isotropic construction, nor proved that isotropy changes the optimal exponent or coefficient.

7.2 Reciprocal electric–magnetic constitutive tensors

With the same incident-amplitude normalization, let

$$\Psi = (E_{\mathrm{phys}}, Z_0 H_{\mathrm{phys}})/E_{\mathrm{ref}}, \qquad q = (P_{\mathrm{phys}}/\varepsilon_0, Z_0 M_{\mathrm{mag}})/E_{\mathrm{ref}} = \chi_6 \Psi.$$

Here M_{mag} is physical magnetization, defined by $B_{\mathrm{phys}} = \mu_0(H_{\mathrm{phys}} + M_{\mathrm{mag}})$. Thus $D_{\mathrm{phys}}/(\varepsilon_0 E_{\mathrm{ref}}) = \Psi_E + q_E$ and $cB_{\mathrm{phys}}/E_{\mathrm{ref}} = \Psi_H + q_M$. Assume the exterior is vacuum, the solution is well posed, $B_6 = \operatorname{Im}_H \chi_6 \succeq 0$, and the reciprocal constitutive relation is

$$\chi_6^T = J\chi_6 J, \qquad J = \operatorname{diag}(I, -I).$$

Suppose a measurable nonnegative η_6 satisfies $|\chi_6 v|^2 \le \eta_6 v^\dagger B_6 v$ for all six-vectors, and $0 < M_6 = \int_D \eta_6 < \infty$.

Proposition 7.1 (Six-component response bound)**.** *For unit incident electric amplitude and opposite time-reversed waves,*

$$S \le \frac{k^3 M_6}{6\pi} A, \qquad X^2 \le 2kM_6 A, \qquad \Gamma \le \frac{k^3 M_6}{6\pi + k^3 M_6} < 1. \tag{37}$$

If also $\eta_6 \le \bar{\eta}_6$ and $D \subset B_a$, then $S \le \bar{\eta}_6 ka\, A$.

Proof. The normalized cross sections are $A = k\int \Psi^\dagger B_6 \Psi$ and $X = k \operatorname{Im}\langle \Psi_{\mathrm{inc}}, q\rangle$. Writing $E = \Psi_E$, elimination of Maxwell's equations gives $(\nabla \times \nabla \times -k^2)E = k^2 q_E + \mathrm{i}k\nabla \times q_M$, hence

$$F(n) = \frac{k^2}{4\pi}\int_D [\Pi_n q_E(x) - n \times q_M(x)] e^{-\mathrm{i}kn\cdot x}\, \mathrm{d}x.$$

For fixed $u, v \in \mathbb{C}^3$,

$$\int_{S^2} |\Pi_n u - n \times v|^2 \, \mathrm{d}\Omega = \frac{8\pi}{3}(|u|^2 + |v|^2).$$

The cross term vanishes because it is odd in n. Minkowski and weighted Cauchy–Schwarz now give the first inequality exactly as in Theorem 3.1. The incident six-vector has squared norm 2, which supplies the factor two in the extinction inequality. Lorentz reciprocity with $\Psi_- = J\overline{\Psi_+}$ equates the forward work and hence the extinctions. Energy conservation and the common-interval argument give (37).

For the radius bound, the six-component angular operator is the electric-current angular operator together with its magnetic copy composed with the tangent rotation $v(n) \mapsto -n \times v(n)$. This rotation exchanges the two vector-harmonic families. Thus the nonzero radiation eigenvalues are $\lambda_\ell^M + \lambda_\ell^N$, each with multiplicity $2(2\ell + 1)$. The sum identity in Theorem 4.2 bounds every such eigenvalue by ka. Use $\|q\|_2^2 \le \bar{\eta}_6 A/k$. □

The electric-only spectrum must not be reused unchanged in a magnetic or bianisotropic calculation. The elementary six-component response framework also has prior precedent [2]. We have not asserted a general six-component rough-coefficient well-posedness theorem under (37)'s response hypothesis.

7.3 A one-sided real cap: a failed inference, not a counterexample

If the only real-part constraint is $\mathrm{Re}_H\, \varepsilon_r \preceq PI$, then the material values

$$\varepsilon_n = (1 - n + \mathrm{i}\delta)I, \qquad n = 1, 2, \ldots, \tag{38}$$

obey that upper cap for $P \geq 1$, the same loss floor δ, and a fixed imaginary upper cap $Q \geq \delta$. Their response factor is $\eta_n = (n^2 + \delta^2)/\delta \to \infty$. They can be realized at ω_0 by passive Drude permittivities

$$\varepsilon_n(\omega) = 1 - \frac{\omega_{p,n}^2}{\omega(\omega + \mathrm{i}\gamma_n)}, \qquad \gamma_n = \frac{\delta\omega_0}{n}, \qquad \omega_{p,n}^2 = n(\omega_0^2 + \gamma_n^2).$$

Substitution gives exactly (38) at ω_0, and the imaginary part is positive for $\omega > 0$.

This family rigorously disproves the inference that the stated one-sided caps bound susceptibility or the loss-weighted response. It does *not* disprove existence of every possible positive F for that enlarged class: a homogeneous isotropic sphere in this family has $\Gamma = 0$. No sequence with fixed R, V, Q, δ, E_0 and $\Gamma \to 1$ in the enlarged class has been established here. Its existence question remains open, with neither a valid universal proof nor a physical counterexample supplied.

7.4 Removing a full tensor loss floor

The integrated condition (12) permits nonuniform and vanishing loss when responding directions remain controlled. It excludes a lossless direction with nonzero susceptibility. Positive trace loss, positive total integrated loss, or positivity only on one polarization does not imply that condition. The earlier planar vector construction exploits precisely such polarization structure; its scalar determinant lower bound cannot be transferred to general tensors.

If one merely fixes size and bounded positive real contrast while allowing arbitrary vanishing loss, the current proof need not have a bounded M. A replacement argument based on compactness would need uniform Maxwell resolvent estimates, a specified topology for the coefficients and geometry, a justified homogenization closure, and uniqueness for the limiting class. Weak convergence of coefficients alone supplies none of these automatically. Uniqueness results for piecewise regular anisotropic coefficients [21] should not be cited as a theorem for an arbitrary enlarged essentially bounded class. We leave that sharper loss-free-resource question unresolved.

7.5 Multiple channels, other backgrounds, and bandwidth

The radiation-efficiency inequality is independent of the incident plane wave once its induced current is given. For a general incident electric field f, the extinction estimate uses $\int_D \eta\, |f|^2$ in place of M. If the selected reciprocal pair has the same positive extracted power under a common normalization, the same interval-width argument applies. Finite-power channel fractions and plane-wave cross sections have different units and cannot be interchanged without that normalization.

In a waveguide, a substrate, or another background, the complete positive radiation operator includes every open and guided channel. A bound on that operator and a compatible loss metric can reproduce the abstract reasoning, but the vacuum constants and vacuum ball spectrum are not automatically valid. Uncounted outgoing channels cannot be called absorption.

All main results here are single-frequency statements. Uniform hypotheses over an interval give corresponding pointwise bounds, but a genuinely sharper bandwidth tradeoff requires constraints on dispersion, oscillator strength, static response, or other causal resources. No finite-bandwidth optimum is proved here.

8 Reproducible computations and certification

The analytical theorems do not depend on an optimizer or a discretized Maxwell model. Computations serve three separate purposes: exact coefficient checking, evaluation of an infinite-dimensional bound, and diagnostics of a physical discretization.

8.1 Certified evaluation of the infinite spectrum

For $0 \le s \le z$, the integral representation of j_ℓ gives $|j_\ell(s)| \le s^\ell/(2\ell+1)!!$. The recurrence $(sj_\ell)' = (\ell+1)j_\ell - sj_{\ell+1}$ then gives

$$\lambda_\ell^M \le \frac{z^{2\ell+3}}{(2\ell+3)((2\ell+1)!!)^2},$$
$$\lambda_\ell^N \le \left[\ell(\ell+1) + \left(\ell+1+\frac{z^2}{2\ell+3}\right)^2\right] \frac{z^{2\ell+1}}{(2\ell+1)((2\ell+1)!!)^2}. \tag{39}$$

For an integer $L \ge 1$ put

$$C_L = 2 + \frac{2z^2}{(L+2)^2} + \frac{z^4}{(2L+5)^2(L+2)^2}, \qquad r_L = \left(\frac{L+3}{L+2}\right)^2 \frac{z^2}{(2L+5)^2}.$$

Combining the positive terms of (39) shows, for $\ell \ge L+1$,

$$(2\ell+1)(\lambda_\ell^M + \lambda_\ell^N) \le C_L \frac{(\ell+1)^2 z^{2\ell+1}}{((2\ell+1)!!)^2}.$$

For clarity, after factoring $(\ell+1)^2$, the leading term is $(2\ell+1)/(\ell+1) \le 2$; the two quadratic-in-z terms together are $(4\ell+3)z^2/((2\ell+3)(\ell+1)^2) \le 2z^2/(L+2)^2$; and the fourth-order term is bounded by that in C_L. Successive majorants have ratio at most r_L. If $r_L < 1$,

$$\sum_{\ell>L} (2\ell+1)(\lambda_\ell^M + \lambda_\ell^N) \le T_L := \frac{C_L (L+2)^2 z^{2L+3}}{((2L+3)!!)^2(1-r_L)}. \tag{40}$$

The omitted D_α and H_α tails are consequently at most $2\pi T_L/(k^3\alpha)$ and $2\pi T_L/(k^3\alpha^2)$. Every omitted radiation eigenvalue is at most $T_L/(2L+3)$. This is a certified remainder estimate, not a physical multipole cutoff.

The script `agents/bound_derivation/certified_ball_bounds.py` uses exact rational alternating series for finite Bessel values and 100-digit `Decimal` interval arithmetic with directed rounding. Its analytic inequalities and the implemented signs were independently audited. NumPy and SciPy choose an initial candidate α; the subsequent certificate does not trust that candidate. An explicit feasible current of the form (29), evaluated at a slightly larger α, gives an upper enclosure of the relaxed minimum; the dual inequality gives a lower enclosure.

The calculations explicitly evaluate $\ell \le 60$, with every higher order included by (40). The largest omitted D_α tail in Table 1 is below 1.1×10^{-76}. All four interval Parseval checks contain zero. These are computational certificates of the formulas, not Lean proofs and not substitutes for the analytical derivations.

ka	k^2X	Certified interval for ρ	Certified Γ upper bound
1	10	[0.1193053312818, 0.1193053312839]	0.5260508725652
3	80	[0.2107489289841, 0.2107489289869]	0.7236130760010
5	200	[0.2416386685242, 0.2416386685268]	0.7211975977659
10	800	[0.3203603686451, 0.3203603686476]	0.6607668654842

Table 1: All-mode certificates for $\bar{\eta} = 10$. The last column uses only $\bar{\eta}\Lambda/(1+\bar{\eta}\Lambda) - \rho$; adding the integrated-budget and cap constraints can improve it. Entries are rounded outward. They bound physical objects but are not physical realizability examples.

8.2 Exact algebra and retarded discrete-dipole diagnostics

The independent script `checks/maxwell_checks.py` verifies the sphere's displayed rational coefficients and tests (13) for 1,000 reproducibly seeded noncommuting real symmetric matrix pairs. It then solves a retarded, radiation-damped discrete-dipole model for both incident waves. For cubic cells of volume h^3 the inverse static polarizability is $(\chi^{-1} + I/3)/h^3$; the self radiation correction is $-\mathrm{i}k^3 I/(6\pi)$. All off-diagonal interactions use the complete retarded dyadic kernel. Absorption is computed from $-\operatorname{Im}_H \chi^{-1}/h^3$, and scattering is independently integrated over the full sphere using transverse fields.

For a cube of side 2 at $k = 1$ with two anisotropic halves and real incident polarization, the results are: Extinction reciprocity and independently integrated radiation-energy balance agree to

Grid per side	Cells	X	Γ
4	64	1.6628669012	0.0607360786
6	216	1.6933783436	0.0593898585
8	512	1.7050488784	0.0589698540
10	1000	1.7105119509	0.0588035214

Table 2: A reciprocal retarded point-dipole discretization. Tensor values, quadrature, mesh construction, residuals, and both illuminations are in the script and JSON output.

approximately 2.4×10^{-15} relative error. The last two asymmetry values differ by about 0.28%. These residuals check the finite dipole system, not a certified continuum discretization error.

The voxel approximation of the sphere illustrates that distinction. At $ka = 0.1$, normalizing by actual voxel material volume gives $\Gamma/(k^3 V_{\mathrm{voxel}}) \approx 0.018558$ and 0.018567 on the two reported grids, versus the exact continuum coefficient $120/(2071\pi) \approx 0.01844384$. The persistent roughly 0.6–0.7% offset is compatible with boundary shape and discretization error. Shrinking a fixed-index voxelization preserves a faceted shape; it does not constitute convergence to a sphere. No continuum sharpness claim rests on these numerical values.

9 Relation to the three preceding manuscripts

The recovered source archives and PDFs were copied into a separate research directory and left unmodified. Their SHA-256 hashes are recorded in `prior_source_checksums.json`. The supplied ordinary proofs were treated as claims to audit. The principal conclusions of the targeted re-examination are as follows.

Previous result	Quantifiers and normalization	Present relation
Restricted scalar multilayers	A sharp inverse-square thickness asymptotic in a specified scalar layered setting with its material and target assumptions.	Preserved. It is not a bound for arbitrary vector compact bodies. The general response inequalities do not recover its sharp coefficient.
Reciprocal channel and planar-vector manuscript	Sharp abstract channel envelopes, including the scalar two-port normalized cap $\sqrt{10\sqrt{5}-22}$; exact laterally infinite vector dielectric constructions.	Port power fractions and finite-power reference extinction differ from free-space cross sections. Additional polarization and channels invalidate a universal extrapolation of the two-port cap.
Compact Maxwell manuscript	At fixed k, $P > 1$, and $Q > 0$, a sequence of compact reciprocal dielectric bodies has $\Gamma \to 1$; each member has finite size and positive loss, but there is no common volume, diameter, or loss floor.	No fatal error was found in the targeted aperture/coercivity audit. The present bounds quantify necessary resource divergence and explain why that sequence does not contradict a fixed-resource gap.

9.1 Audit of the compact localization proof

The key analytic steps reviewed were the fixed-stack coercivity, distributional contact term, finite-aperture localization, treatment of the grazing light circle, ordered positive-loss regularization, and recovery of total scattering through $X - A$. The Weyl fiber singularity is locally integrable in L^1 near grazing, but not globally square integrable there. The proof uses the appropriate mixed estimates rather than asserting the false global L^2 bound. Smooth aperture windows and rapidly decreasing Fourier tails supply the stated localization control. The stack and its loss floor are fixed before the aperture limit is taken. Constants are not uniform when loss vanishes or thickness grows.

No fatal proof error was identified in those reviewed steps. This finding is an AI-led mathematical audit, not independent human peer review or a formalized Maxwell proof. The prior supremum conclusion remains a research result with priority uncertain. Conditional on its ordinary proof as audited, it resolves the unrestricted compact-object version of the normalized problem: $\sup \Gamma = 1$, while each positive-extinction compact object has $\Gamma < 1$ by continuity of its far field and the forward optical theorem. It does not give a fixed-resource optimum.

9.2 Correction to the formal-verification scope

The recovered `OQ7_Lean4_verified_supplement/README.md` states that it contains **43 explicitly authored mathematical theorems**. It also explicitly states that it does **not** prove the physical thickness theorem identified there as Theorem 5.1. The generalized manuscript's `formal_correspondence.tex` separately lists the physical thickness asymptotic as unverified in Lean. The recovered release reports no declaration `OQ7.theorem51` and no recovered 692-theorem project.

Accordingly, the claim that the entire restricted physical thickness theorem is fully Lean-verified could not be corroborated from these available files. The verified algebraic subset should not be relabeled as a formal proof of its missing Maxwell, material, and asymptotic premises. This is a statement about the recovered release; it does not establish that a different, later archive cannot

exist. No prior Lean project or log was changed, and no new Lean formalization was undertaken for the present theorem-discovery phase.

10 Targeted novelty review

The review used actual primary publications and author manuscripts, with precise access limitations recorded in `agents/literature/targeted_novelty_review.md`. It is a targeted comparison, not an exhaustive proof of priority.

Prior publication	Closest established ingredient	Consequence for positioning
Miller et al. (2016) [2]	Geometry-independent absorption/scattering bounds and the loss-weighted susceptibility figure of merit, including general tensor electric–magnetic formulations.	Neither the response budget concept nor its tensor/volume extension should be advertised as a new general principle.
Schab et al. (2020) [3]	Prescribed-illumination absorption–scattering tradeoffs; the radiation-to-loss eigenvalue bounds scattering efficiency.	The elementary Γ bound is an OQ7 consequence of an established efficiency inequality. This is the closest antecedent.
Gustafsson et al. (2020); Kuang et al. (2020) [4, 5]	Full-current power constraints, spherical radiation modes, and single-frequency material/size response bounds.	The operator framework and sphere decomposition require attribution. The explicit all-order linear estimate and ensuing paired bound should be compared at the formula level.
Molesky et al. (2020, 2022); Shim et al. (2024) [14, 15, 16]	T-operator constraints, local conservation, and cross-source constraints enforcing a common structure.	The present separate-current relaxation omits exactly the same-device compatibility needed for physical joint sharpness.
Gustafsson–Capek (2019); Andersen–Frandsen (2005) [7, 8]	Directivity, gain, receiving-antenna absorption, and efficiency costs.	Forward-directivity reasoning is established. The current norm and infinite tail must remain controlled to address superdirectivity.
Hopkins et al. (2015, 2016) [10, 9]	Reciprocal absorption/scattering dichroism and nonorthogonal modes; anisotropic or inhomogeneous lossy media can exhibit this mechanism even without retarded coupling.	The qualitative mechanism of the tensor sphere is not new. Its role here is a complete Maxwell proof of a scaling exponent.
Sounas–Alù (2014, 2017); Ma et al. (2025) [11, 12, 13]	Extinction symmetry, time-reversal field constraints, loss-channel extensions, and weak-loss asymmetry results.	Different loss and normalization hypotheses must remain distinct; one cannot apply a lossless field theorem unchanged to all lossy bodies.

Prior publication	Closest established ingredient	Consequence for positioning
Gustafsson (2024); Kuang et al. (2025) [6, 17]	Modal bounds and synthesis, and geometry-dependent vector-wave channel spectra without arbitrary finite-channel cutoffs.	Optimized incoming waves or a different propagation operator do not by themselves give this fixed plane-wave-pair optimum.
Wang et al. (2018); Máñez-Espina et al. (2026) [18, 19]	Angularly asymmetric metasurfaces and a two-mode model of directional absorption in reciprocal bianisotropic metasurfaces.	Same-side opposite angles, periodic structures, and model-specific two-mode limits are not identical to arbitrary compact time-reversed plane-wave scattering.

A particularly important unresolved priority lead is Mann and Alù's 2023 conference contribution, *Maximum Asymmetric Absorption and Scattering in Bianisotropic Particles* [20]. Its official abstract discusses dipolar bianisotropic conditions for critical coupling under one illumination and zero absorption under the opposite illumination. Only the official abstract was recovered; the complete derivation and any subsequent journal version were not. Its normalization, resource assumptions, attainable tensors, and relation to the earlier compact manuscript's dipolar claims must be checked before asserting originality. An abstract is sufficient to identify a serious priority lead, but insufficient to attribute an unexamined theorem to it.

The plausible contribution of this continuation is therefore the precisely quantified compact-Maxwell OQ7 resource theorem, the uniform all-order size estimate, the exact positive-extinction current minimum with its explicit proof and infinite-tail certificates, and the physical small-volume scaling theorem. Some or all may be reformulations or consequences of earlier optimization frameworks. No claim of established scientific novelty is made.

11 Proof ledger and OQ7 verdict

Claim	Status and remaining obligation
Positive gap for bounded response; explicit P, Q, δ, V, R theorem	Complete ordinary proof above. Full tensor loss supplies well-posedness for the essentially bounded capped class. Independent AI adversarial rederivations found no fatal error.
All-order $\Lambda(ka) \le ka$ and $\Lambda = \Theta(ka)$ at large size	Complete proof for the radiation operator. The latter statement is operator-order sharpness only.
Positive scattering floor and exact ρ	Complete proof and explicit optimizer for the exact stated current relaxation. Physical constitutive realizability is not proved.
Strongest physical Γ bound at fixed resources	Equations (31) and (34) are valid upper bounds. Equality, optimal coefficients, and joint two-illumination attainment are unresolved.
Small-volume exponent	Exact-Maxwell anisotropic sphere proves order k^3V in the allowed circular time-reversal class. Isotropic/real-linear matching and a fixed-positive-extinction analogue remain open.

Claim	Status and remaining obligation
One-sided real cap with unbounded negative contrast	Response-budget inference is false. The proposed asymmetry theorem for that enlarged class is unresolved; no physical counterexample approaching one was proved.
Lossless responding sectors and weaker loss assumptions	Not covered unless a finite response inequality remains available. Uniform resolvent/closure or a physical counterexample is required.
General constitutive media and other backgrounds	Elementary reciprocal six-component bound and vacuum linear-radius bound proved for well-posed solutions. Other backgrounds need their complete radiation operator and appropriate normalization.
Finite bandwidth and causality-limited optimum	No new bandwidth optimum. The single-frequency tensor examples are compatible with passive causal dispersion.
Novelty and formal/human verification	Primary-source review completed with the Mann–Alù access gap explicit. No priority certification, complete Lean proof, or independent human peer review is claimed.

The fixed-resource existence question is answered affirmatively for a broad, physically explicit class, without a minimum feature size or multipole cutoff. An arbitrarily small deficit $1 - \Gamma$ forces divergence of the integrated response and of the product $\bar{\eta}ka$ under the derived necessary-cost inequalities. At positive extinction the exact radiation resolvent yields an additional obstruction. These statements make the resource-free supremum compatible with a quantitative physical cost.

They do not identify the sharp resource-constrained supremum. The main next mathematical task is a joint optimization for the two currents that enforces the same local reciprocal constitutive operator, including the reactive Maxwell constraints, followed by a physical construction or a proof of a residual gap. Cross-source T-operator methods provide a concrete prior framework for that step [15, 16]. The absence of a lower real-part bound is a separate theorem-or- counterexample problem and should not be silently folded into the proved capped class. The original OQ7 is thus substantially narrowed, but not fully closed across its resource, constitutive, normalization, and bandwidth interpretations.

A Reproduction and preserved evidence

The accompanying archive contains this standalone LaTeX source, the compiled PDF, the two numerical scripts and their JSON outputs, the independent derivation/audit notes, the targeted literature review, and the original-file checksum inventory. The earlier manuscripts and archives themselves remain unchanged and are not duplicated into the new research release. External publications are referenced by their primary URLs rather than redistributed as full texts.

To compile the manuscript, run `pdflatex` twice on the supplied `.tex` file. It has an embedded bibliography and no external figure or data-file dependency. To reproduce computations from the archive root:

```
python agents/bound_derivation/certified_ball_bounds.py
python checks/maxwell_checks.py
```

The first script writes its certificates next to itself. The second writes `results/maxwell_checks.json`. The interval arithmetic uses Python's standard library; NumPy/SciPy are used for candidate

selection and the independent discrete-dipole computation. Exact dependency versions, instructions, and scope distinctions appear in `README.txt` and `verification/release_checks.json`. Numerical reruns are not required to validate the ordinary proofs.

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